holography does not compromise on resolution. When you look at an object in the physical world, you perceive it due to how light reflects off it and reconstructs it in front of your eyes. Holography replicates this process by capturing the unique light reflection patterns and using a laser to project them, creating a lifelike three-dimensional image.

This method provides genuine depth perception, making it an attractive choice for various applications, including heads-up displays. Holographic displays can deliver bright, true, three-dimensional visuals, enhancing drivers' understanding of their surroundings. However, holography is computationally intensive, consumes significant power, and currently needs to improve image quality compared to other alternatives.

Regarding heads-up displays, spatial light field displays and holography are under consideration. However, the choice between the two depends on the specific application and requirements. Light field displays are advancing more rapidly for consumer electronics and panel side displays while promising holography might take a bit longer to mature, especially for commercial use.

Vision beyond the dashboard

The automotive world is currently undergoing a fascinating evolution in dashboard displays and heads-up displays (HUDs). Although these two technologies coexist, a transformative shift is on the horizon.

Dashboard displays and HUDs are distinct elements within a vehicle's user interface. However, the future of automotive displays holds the potential for a significant transformation. While dashboard displays might eventually become obsolete, the road to this scenario is marked by various adaptations and challenges.

HUDs is a particularly intriguing development. Positioned ahead of the driver and projected onto the windshield, the HUDs are paving the way for future augmented reality (AR) displays. Their primary intent is to significantly reduce distractions. In current setups, drivers often divert their gaze to interact with GPS, adjust the radio, or manage phone calls, usually on the side panel. The HUDs aim to curtail this by providing a translucent display that allows the driver to access necessary information without shifting their focus from the road. Such distractions have been responsible for numerous accidents, endangering not just the vehicle's occupants but also pedestrians and other road users.

The concept of a future where the traditional dashboard display is replaced entirely by a heads-up display is intriguing. Imagine a scenario where HUDs prove so successful that they can replicate and surpass everything the dashboard display offers. HUDs make this advancement possible by projecting translucent images that provide a superior user experience. In the future, drivers might no longer require the conventional dashboard display. However, it's crucial to acknowledge that we are not yet at that point. There are ongoing adaptations in the automotive industry, and the coexistence of dashboard displays and HUDs is a testament to the transitional phase.

One significant challenge facing HUDs is their performance under varying lighting conditions. Bright outdoor environments, for example, can make it difficult to see information displayed on screens, similar to the struggle we encounter when viewing our smartphones on sunny days. The effectiveness of HUDs heavily relies on their ability to combat issues related to incident light, ensuring that crucial information remains visible to the driver.

HUDs themselves can employ different technologies, such as holography and LCDs. Holographic HUDs are known for their exceptional brightness, which can mitigate visibility problems caused by intense ambient light. On the other hand, LCD-based HUDs may need help delivering clear visuals under extremely bright conditions, making them less favourable in specific scenarios.

The automotive display industry is transforming remarkably, driven by technological advancements and the pursuit of safer and more immersive driving experiences. From the dominance of LCD and OLED technologies to the promising emergence of microLEDs, the future of automotive displays looks brighter than ever. The development of light field displays and computer-generated holography introduces the possibility of three-dimensional realms within vehicles, redefining how we interact with in-car displays. Given its manufacturing capabilities and expertise in micro-LEDs, India's potential contribution to this evolving landscape is undeniable.

The competition between traditional dashboard displays and the HUDs hints at a future where HUDs might replace conventional dashboards, provided they can address challenges related to visibility in varying lighting conditions. As the automotive display industry continues to evolve, it promises to deliver safer, more engaging, and technologically advanced driving experiences for all. Moreover, the industry's trajectory points toward larger automotive displays, a global trend that enhances the driving experience and increases revenue, reaffirming the exciting future of automotive displays. EFY

The co-author, Nidhi Agarwal, works as Technology Journalist at EFY